

ALGEBRA 1

- 1) A lab assistant mixes 6 L of a 25% saline solution with 4 L of a 50% saline solution. What is the percent concentration of the new solution?
- 2) A coffee shop blends 12 lbs of beans that are 80% Arabica with 8 lbs of beans that are 60% Arabica. What percent of the new blend is Arabica?
- 3) A chemist combines 5 mL of a 40% acid solution with 15 mL of an 8% acid solution. What is the acid percent of the resulting mixture?

4) A juice maker mixes 3 L of juice that is 45% real fruit with 7 L of juice that is 15% real fruit. What percent of the final juice is real fruit?

5) A jeweler melts 20 g of an alloy that is 75% gold with 30 g of an alloy that is 25% gold. What is the gold percent of the new alloy?

Part B · Percent Change (5 problems)

Directions: Find each percent change. State if it is an **increase** or a **decrease**. Round to the nearest tenth of a percent if needed.

6) From 50 to 65 → _____ %, _____ (I / D)

7) From 120 to 90 → _____ %, _____ (I / D)

8) From 25 to 40 → _____ %, _____ (I / D)

9) From 200 to 170 → _____ %, _____ (I / D)

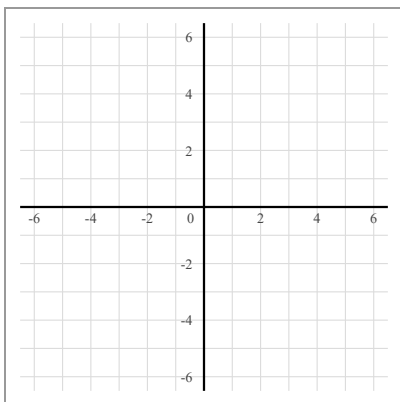
10) From 18 to 9 → _____ %, _____ (I / D)

Part C · Sketch the Graph of Each Line (5 problems)

Directions: Find both intercepts (or use slope-intercept form), plot at least two points, and draw the line on the grid. Label the line with its equation.

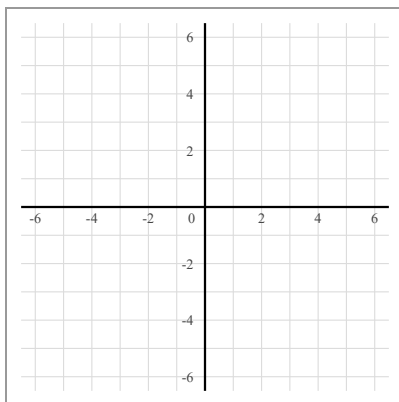
11)

$$2x + 3y = 6$$



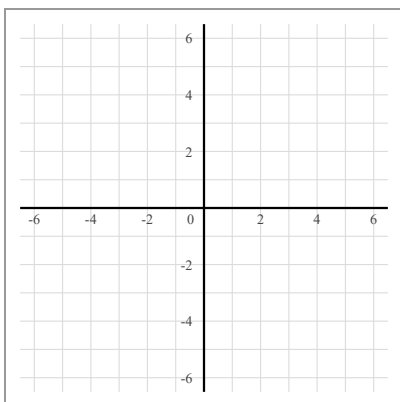
12)

$$x - 4y = 4$$



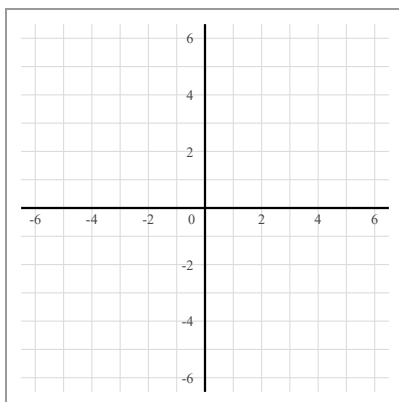
13)

$$5x + 2y = 10$$



14)

$$6x - 3y = 0$$



15)

$$y = \frac{1}{2}x - 3$$

